

NatWest

# How to Build Your Robot



# What you'll need

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- A **Camjam robotics kit**, including:
  - The box 'chassis'
  - Two motors
  - One motor controller board
  - Two wheels
  - Battery box & 4 AA batteries
  - Roller ball wheel
- **Raspberry Pi Model 3A computer** and peripherals:
  - USB Power bank
  - Wireless Keypad
  - HDMI Cable & Capture device
- As well as:
  - Scissors
  - A screwdriver
  - Double sided foam tape
  - A tape measure
  - Lots of arts and crafts...
- **Come up with a name for your robot and write it here:**  

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# A thought...

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“What’s really angering about instructions of this sort is that they imply there’s only one way to put this thing together - their way. And that presumption wipes out all the creativity. Actually there are hundreds of ways to put the thing together and when they make you follow just one way ... you lose feeling for the work. And not only that, it’s very unlikely that they’ve told you the best way.”

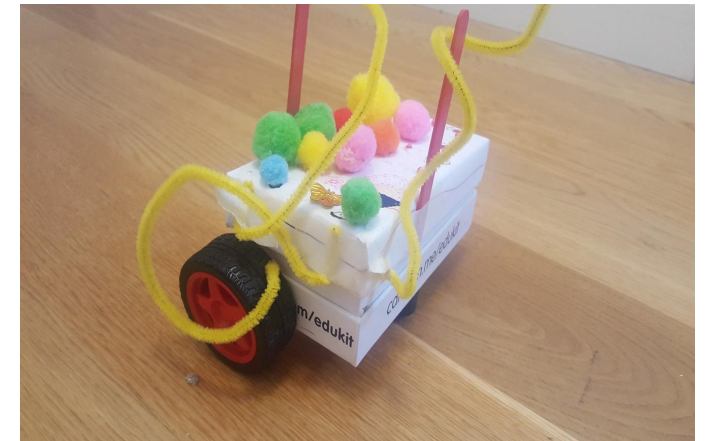
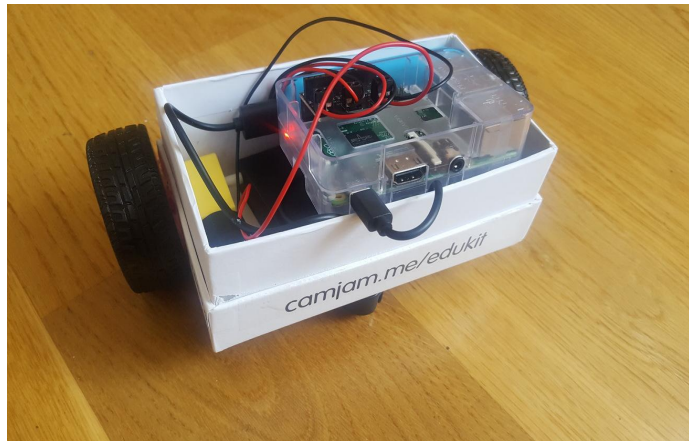
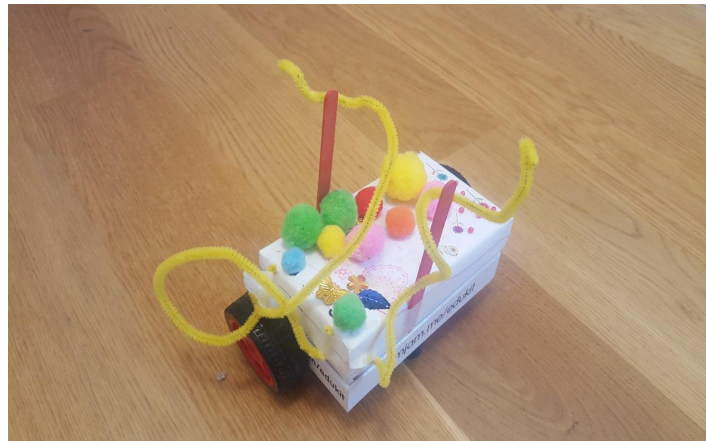
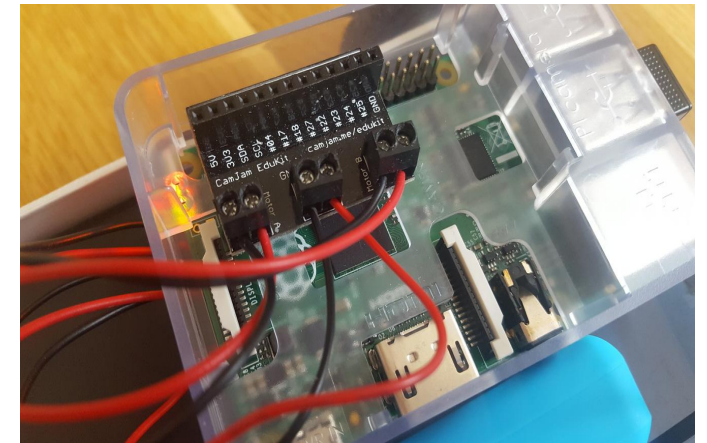
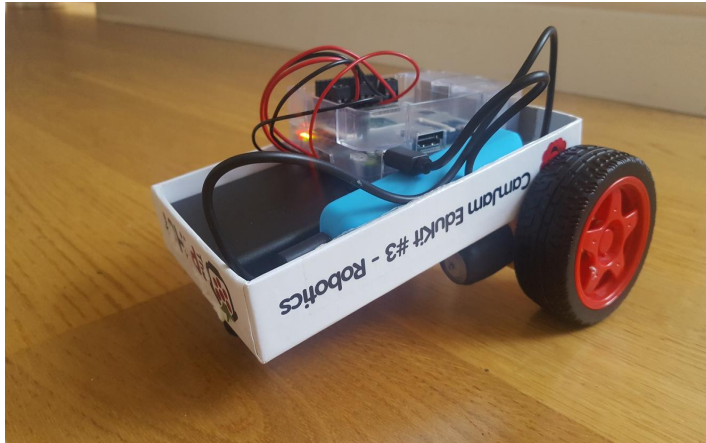
- Robert M Pirsig  
*Zen and the art of Motorcycle Maintenance*

*(Meaning: we won’t tell you how to put the robot together, you have to figure out the best way yourselves. Think about strength, sturdiness and weight distribution!)*



# What it might look like

Here are some pictures of a basic assembled robot...





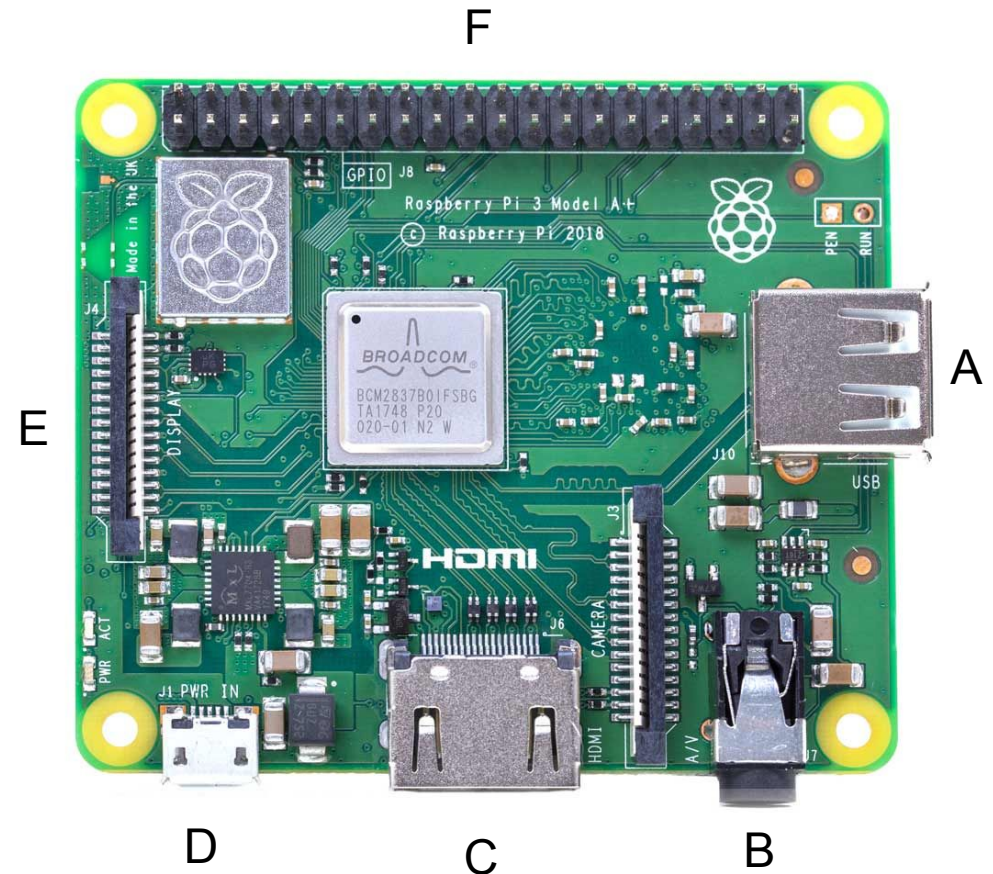
# About the Raspberry Pi 3A

It's a fully fledged £24 PC that can do whatever a regular PC can do.  
But it's also great for controlling robots...

It has the following connectors:

- A: 1 USB port
- B: 3.5mm audio output
- C: HDMI output (for video & audio)
- D: micro-USB 5v power input
- E: micro-SD memory card slot (underneath)
- F: GPIO port.

The latter is the “General Purpose IO” port, and we’re going to have a lot of fun with this today. Most PCs don’t have them, and it’s part of what makes the Raspberry Pi so fun and flexible!

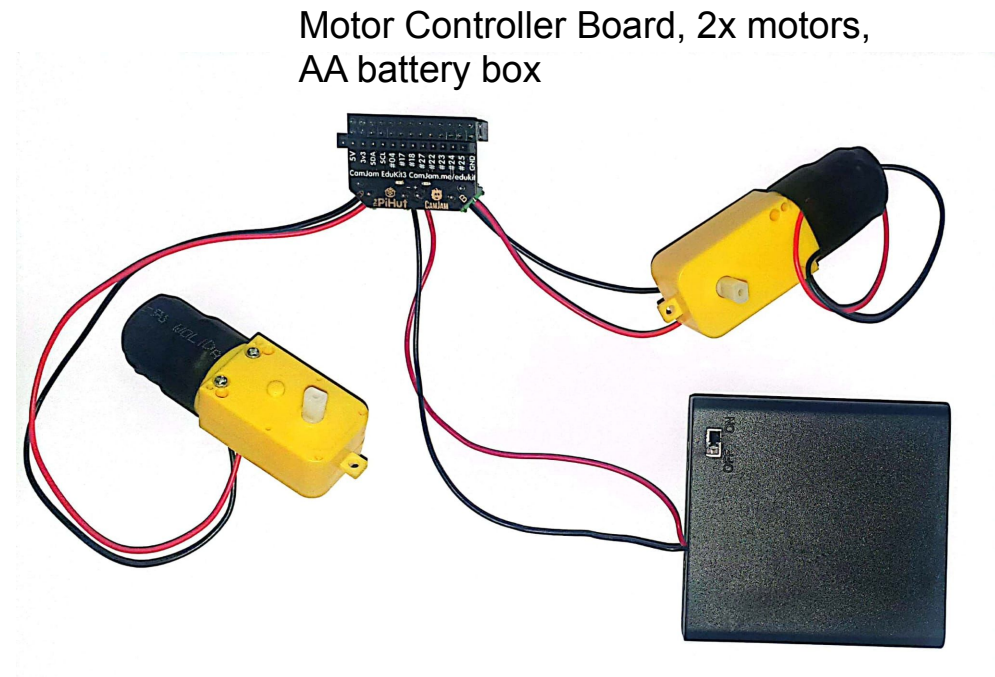




# Getting Started

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- Empty the box and decide which is the top/front. Think about how to attach the motors, wheels and the third ball-bearing wheel for maximum strength and stability. It will be a three-wheeled robot.
- Think about how the powerbank, Raspberry Pi and motor battery box will go inside the robot. Don't forget to put the AA batteries in.
- The motor controller board connects to the Raspberry Pi, but you need to connect the motor wires first - make holes through the box for them.
- Think about where all the wires will go, then assemble the parts and attach everything to the chassis.





# Firing up the Robot

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Connect up your Raspberry Pi and boot it up:

- Connect the HDMI output to the laptop using the HDMI cable and HDMI capture device
- Connect the wireless keyboard/mouse using the dongle you will find in the battery compartment on the back of it (don't mix these up with other teams - each pair is unique)
- Ensure the switch on the motor battery box is set to "on"
- Prop up the robot to ensure the wheels are not touching the desk
- Connect the powerbank to the Raspberry Pi last.
- Go to <https://nerdzap.com/view/> on your laptop and click "start", make full screen. You will not use the keyboard/mouse on the laptop, you will use the wireless controller only, which should be switched on..
- It should boot up, if not seek help.



# Running the Robot Program

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To test your robot:

- Type `ls` to show the files in the current folder (don't type `sl` by accident!)
- At the prompt type `./init.sh` and hit 'enter', it should say "Success". This resets everything ready for you.
- At the prompt type `cd robot` hit 'enter'
- Then type `py robot.py` and hit 'enter'
- Wait for "Ready for Robot Commands..." to appear
- Press the `t` key on your keyboard (is your motor battery box switched on?)
- Press `esc` to stop the robot program
- Type `nano robot.py` to edit the program, press `ctrl-s` to save and `ctrl-x` to exit

Let's look at those challenges:

Skim read through the first half of `robot.py` (hint: read the comments, which are red)

Run it again. This time, press `1` on your keyboard. Can you figure out what is happening?



# What's going on?

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Making the robot move around is relatively simple:

- The two motors are connected to the pins of the GPIO port.
  - All the pins are initially off ('0' state)
  - Turning on the right pins ('1' state) makes the motors run.
  - Making both motors run in the same direction will make the robot go straight.
  - Making the motors run in opposite directions will make the robot turn.
- We're using Python code to control the pins. Python is a programming language which makes this quite easy.
- You're going to write some Python to compete in the challenges.



# The Challenges

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Four rounds of challenges:

## 1. Distance Challenge

drive *exactly* 1m in distance (helps calibrate your robot too!)

## 2. Obstacle Avoidance

drive around an obstacle without hitting it and without touching the edges

## 3. Aesthetic Design

judged by the Headteacher for lots of extra points

## 4. Arena Battle

A battle for glory in the Arena of Robot Carnage!



# Basic Robot Commands and Python

|                                                              |                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Basic Robot Movement</b>                                  | <p>Comments have a # at the start, and show up in red. Read them to find out what something does. Write them to remind yourself.</p>                                                                                                                                                                                   | Here are some examples to give you an idea.                                                                                                                                                                                                                                                                                                                           |
| <b>Forward(n)</b><br><i>-- run forward for n seconds</i>     | <p># this is a python comment</p> <p>Python functions are neat ways to bundle a set of commands and use them when you want:</p> <pre>def doSomething():<br/>    print("this is what I do")</pre>                                                                                                                       | <pre>def runChallengeA():<br/>    print("going 2 metres")<br/>    # X is time it takes bot travel 1m<br/>    x = 0.4<br/>    # travel 2 metres forward...<br/>    robot.forward(x * 2)</pre>                                                                                                                                                                          |
| <b>Backwards(n)</b><br><i>-- run backwards for n seconds</i> | <p>Whitespace is <i>important</i> in python. Indents (4 spaces or a tab) form the 'blocks' of code (e.g. in a loop, a function or a if-statement). Ask for help if you have trouble.</p> <pre>def myFunction():<br/>    print("this is in my function")<br/>    print("so is this")<br/>    print("this is not")</pre> | <pre>def runChallengeB():<br/>    x = 0.4<br/>    # t is time it takes to turn 90 degrees<br/>    t = 0.6<br/><br/>    # travel 2 metres forward...<br/>    robot.Forward(x * 2)<br/>    # do a little dance...<br/>    robot.Right(t)<br/>    robot.Forward(x)<br/>    # turn around 180 degrees...<br/>    robot.Left(t * 2)<br/>    robot.Backwards(x * 1.2)</pre> |
| <b>Right(n)</b><br><i>-- turn right for n seconds</i>        |                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Left(n)</b><br><i>-- turn left for n seconds</i>          | <p>Variables can store useful stuff to use later:</p> <pre>x = 1.3</pre> <p>You can use these in your commands, e.g. as a multiplier</p> <pre>robot.Forward(2 * x)</pre>                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                       |
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